

Understanding SVM Kernels: How Kernel Choice Shapes Decision Boundaries

1. Introduction

Machine learning models are widely used across many domains such as healthcare, finance, and image recognition. A common task in machine learning is classification, where the goal is to assign data points to categories. However, many real-world datasets are not linearly separable, meaning that a straight line (or hyperplane) cannot effectively divide the classes. Support Vector Machines (SVMs) are powerful supervised learning algorithms that address this challenge. They aim to find an optimal boundary that maximises the margin between classes. While basic SVMs work well for linearly separable data, their true strength lies in their ability to handle complex, non-linear datasets through kernel functions. This tutorial explores how different kernel functions—linear, polynomial, and radial basis function (RBF)—affect the behaviour of SVMs. Using visual experiments and quantitative evaluation, we demonstrate how kernel choice influences model performance and decision boundaries.

2. What is a Support Vector Machine?

A Support Vector Machine is a supervised learning algorithm used for classification and regression. Its objective is to find the optimal hyperplane that separates data into different classes. The optimal hyperplane is defined as the one that maximises the margin between the closest points of different classes, known as support vectors. The margin is important because a larger margin typically leads to better generalisation on unseen data. In two-dimensional space, the hyperplane is simply a line, but in higher dimensions, it becomes a plane or hyperplane. SVMs are effective for linearly separable data, but many datasets are more complex. In such cases, SVM uses kernel functions to transform the data into a higher-dimensional space where separation becomes possible.

3. The Kernel Trick

The kernel trick is a method used by SVMs to handle non-linearly separable data. Instead of explicitly transforming data into higher dimensions, kernels compute the relationships between points as if the transformation has already been applied. Common kernels include:

- Linear Kernel: Suitable for linearly separable data.
- Polynomial Kernel: Introduces curved boundaries based on degree.
- RBF Kernel: Highly flexible and capable of modelling complex patterns.

By using kernels, SVMs can efficiently create complex decision boundaries without increasing computational cost significantly.

4. Experimental Setup

To demonstrate the impact of kernels, we use a synthetic dataset generated using the moons function from `sklearn`. This dataset is non-linearly separable, making it ideal for comparing kernel performance. We split the dataset into training and testing sets to evaluate generalisation performance. Three models are trained using linear, polynomial, and RBF kernels. The same dataset is used for all models to ensure a fair comparison. Decision boundaries are visualised to

provide intuitive understanding of how each kernel behaves.

5. Results and Analysis

The linear kernel produces a straight decision boundary and struggles to classify the data correctly, demonstrating underfitting. The polynomial kernel introduces curvature, improving classification performance. However, increasing the degree too much may lead to overfitting. The RBF kernel produces highly flexible decision boundaries and achieves the best performance on this dataset. It effectively captures complex patterns but must be carefully tuned to avoid overfitting.

5.4 Hyperparameter Effects

Hyperparameters such as C and γ significantly influence SVM performance. C controls the trade-off between margin size and classification error. A small C results in a wider margin but allows more misclassifications, while a large C reduces errors but may overfit. γ determines how far the influence of a single data point extends. A low γ creates smoother boundaries, while a high γ creates highly complex boundaries that may overfit.

5.5 Quantitative Evaluation

Accuracy scores confirm the visual observations. The RBF kernel achieves the highest accuracy, followed by the polynomial kernel, while the linear kernel performs worst. This demonstrates the importance of choosing the right kernel and tuning hyperparameters appropriately.

6. Key Insights

Kernel choice directly affects model performance. Linear kernels are best for simple data, polynomial kernels provide moderate flexibility, and RBF kernels handle complex datasets. There is always a trade-off between underfitting and overfitting, and careful tuning is required to achieve optimal results.

6.1 Limitations

SVMs can be computationally expensive for large datasets. They require careful tuning of hyperparameters and are less interpretable than simpler models.

6.2 Ethical Considerations

Machine learning models can introduce bias if trained on biased data. SVMs are not easily interpretable, which can affect trust and fairness in decision-making systems.

7. Conclusion

SVMs are powerful tools for classification tasks, especially when combined with kernel functions. Understanding how kernels affect decision boundaries allows practitioners to choose appropriate models for their data.

8. References

Cortes, C., Vapnik, V. (1995). Support-vector networks. Scikit-learn documentation: <https://scikit-learn.org> Murphy, K. (2012). Machine Learning: A Probabilistic Perspective.